

SHOPWISE

E-Commerce Conversion Intelligence

IBM Internship Project — Data Analytics + Optional Machine Learning

Student: [Student Name]

Branch / Semester: IT / 7th Semester

1. Executive Summary

SHOPWISE is an end-to-end analytics project that converts online shopping-session data into business KPIs, behavioral insights, and decision-oriented actions. The project follows the Data → Information → Insight → Decision → Action structure covered in the internship session.

The project uses the UCI Online Shoppers Purchasing Intention Dataset. The public dataset contains 12,330 shopping sessions and records whether each session resulted in revenue. The official UCI description identifies the dataset as a business dataset containing browsing, engagement, traffic, visitor, month, weekend and page-value attributes.

2. Business Problem

E-commerce teams receive large volumes of session-level behavioral data but need a compact way to identify conversion patterns and prioritize actions. SHOPWISE addresses this by answering: Which session characteristics are associated with successful purchases, and where can the business intervene to improve conversion?

3. Objectives

- Measure overall conversion performance.
- Compare conversion across visitor types, months and traffic sources.
- Create engagement-oriented derived features.
- Identify patterns that can be translated into business actions.
- Provide an optional predictive layer for purchase intention.

4. Dataset

Source: UCI Online Shoppers Purchasing Intention Dataset, ID 468. The dataset contains 12,330 sessions with 17 input attributes and the Revenue target. The UCI page describes it as a business dataset and provides the official dataset citation and CC BY 4.0 license.

Field group	Examples
Browsing	Administrative, Informational, ProductRelated
Engagement	Page durations, BounceRates, ExitRates
Intent signal	PageValues, SpecialDay
Context	Month, VisitorType, Weekend, TrafficType
Target	Revenue

5. Methodology

The Python pipeline performs data loading, cleaning, feature engineering, KPI calculation, segmentation, visualization and optional classification. Boolean fields are normalized. TotalPages, TotalDuration, AvgSecondsPerPage and EngagementScore are derived to create business-friendly measures.

For the optional predictive layer, the pipeline uses a stratified 80/20 train-test split with random seed 42 and compares Logistic Regression with Random Forest. The evaluation includes accuracy, precision, recall, F1-score

and ROC-AUC. The exact metrics are generated at runtime and written to `outputs/model_metrics.csv`; no model score is fabricated in this report.

6. KPI Framework

KPI	Purpose
Sessions	Measure traffic volume.
Purchasers	Measure successful purchase sessions.
Conversion Rate	Purchasers divided by sessions.
Average Page Value	Summarize the page-value signal.
Visitor Conversion	Compare new and returning visitors.
Monthly Conversion	Detect temporal variation.
Traffic Conversion	Compare traffic-source performance.

7. Insight-to-Action Framework

The project is intentionally designed so that analysis leads to decisions rather than ending at charts. High-exit or high-bounce patterns can trigger UX review; high-intent segments can receive targeted interventions; new and returning visitors can be treated differently; and month/traffic patterns can inform campaign allocation. The final actions should be selected from the executed outputs rather than assumed in advance.

8. Reproducibility

The complete implementation is contained in one Python file. Install the listed dependencies, run `shopwise_analysis.py`, and the script downloads the public dataset and generates KPI tables, CSV outputs and charts under the `outputs` directory.

9. Limitations

The dataset represents historical online-shopping sessions and does not contain full business context such as product margin, campaign cost, customer lifetime value or inventory. Therefore, recommendations should be treated as analytical decision support rather than causal proof. The optional classifier indicates predictive association, not causation.

10. Conclusion

SHOPWISE demonstrates a complete analytics workflow from raw session data to business KPIs, segmented performance views, actionable decision areas and an optional predictive layer. It is intentionally compact enough to be reproducible while still showing the internship's core expectation: converting data into information, insight, decision and action.

Dataset reference

Sakar, C. O. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository. DOI: 10.24432/C5F88Q.

Official dataset page: <https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset>